Credit Card Fraud Detection Dataset Documentation

Overview

This document contains information about the dataset, input features, and sample transaction data used in the Credit Card Fraud Detection project.

Dataset Used

Credit Card Fraud Detection Dataset

The model was trained using a publicly available dataset containing anonymized credit card transaction records.

Dataset Information

- Total Transactions: 284,807
- Fraudulent Transactions: 492
- Normal Transactions: 284,315
- Total Features: 31
- Target Column: Class

Target Labels

- 0 → Normal Transaction
- 1 → Fraudulent Transaction

Important Dataset Note

The dataset uses PCA-transformed anonymized features.

The columns V1 to V28 are transformed numerical variables generated using Principal Component Analysis (PCA).

These are not real-world fields like:

- card number
- customer name
- bank location

The transformation protects sensitive banking information.

Features Used in the Application

Input fields used in the web application:

- Time
- V1 to V28
- Amount

Input Instructions

Users should enter:

- numerical transaction values
- anomaly-related feature values
- transaction amount

The model analyzes the transaction pattern and predicts whether the transaction is:

- Normal
- Fraudulent

Important Input Note

The values V1 to V28 are PCA-transformed numerical features.

Because fraud transactions are extremely rare, random inputs may still predict:

Normal Transaction

Sample Fraudulent Transaction Data

Use the following values to test fraud prediction:

Time = 406

V1 = -2.312227 V2 = 1.951992 V3 = -1.609851 V4 = 3.997906 V5 = -0.522188 V6 = -1.426545 V7 = -2.537387 V8 = 1.391657 V9 = -2.770089 V10 = -2.772272 V11 = 3.202033 V12 = -2.899907 V13 = -0.595222 V14 = -4.289254 V15 = 0.389724 V16 = -1.140747 V17 = -2.830056 V18 = -0.016822 V19 = 0.416956 V20 = 0.126911 V21 = 0.517232 V22 = -0.035049 V23 = -0.465211 V24 = 0.320198 V25 = 0.044519 V26 = 0.177840 V27 = 0.261145 V28 = -0.143276

Amount = 0.00

Expected Prediction

Expected output:

Fraudulent Transaction Detected

Live Web Application

https://anomalydetection-qxwfshrb2rrqsygnv84njz.streamlit.app/

Conclusion

This project demonstrates how machine learning, anomaly detection, and statistical analysis can be used to identify suspicious financial transactions and improve fraud detection systems.